

SENG1050

Lab 7

In this lab, you will gain experience in using PHP to write templates and handle XML data.

[Task 1*](#) - [Task 2*](#) - [Task 3](#)

Task 1 - Simple PHP Template

In this task, you are going to create your own PHP template, and apply it to your website.

1. Follow the steps in Lab 6 to start Apache using the XAMPP Control Panel.
2. Navigate to the virtual host you created in Lab 6, i.e., the `seng1050` directory.

Create a new directory called `theme`.

Within the `theme` directory, create a new file called `template.php` and copy the following code into it:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title><?php echo $title; ?></title>
    <link rel="stylesheet" type="text/css" href="<?php echo $themefolder ?>/style.css" >
  </head>
  <body>
    <?php echo $content; ?>
  </body>
</html>
```

3. Within the same `theme` directory, create a new file called `style.css` and copy the following code into it:

```
body {color: orange}
h1 {color: black}
h2 {color: blue}
h3 {color: red}
h4 {color: purple}
```

4. Return to the `seng1050` directory, create a new file called `index.php` (or open an existing one) and copy the following code into it:

```
<?php
$title = 'First PHP Webpage';
$datafolder = 'data';
$themefolder = 'theme';
$content = '
<h1>First PHP Webpage</h1>
<h2>h2 Heading Here</h2>
<h3>h3 Heading Here</h3>
<h4>h4 Heading Here</h4>
<p>SENG1050 just rocks!</p>
';
?>

<?php include($themefolder . '/template.php'); ?>
```

If you are working with an existing `index.php` file, remove all the existing code and keep it elsewhere for future reference.

5. Open your browser and type `http://seng1050.com` in the URL bar.
You will now see your first webpage which is created using a PHP template.
6. Right click on the page and select 'View Page Source', copy the code into the [HTML validator](#) and see if the code is valid.
If you find any problems in the validator, you should fix them.

Assessment 3 Milestone 1 (Important!)

Once you have completed Task 1 above, you are able to complete milestone 1 of Assessment 3. Please refer to the Assessment 3 documentation and follow the steps listed carefully.

Upon completion of milestone 1, approach your lab instructor to claim 1% of your Assessment 3 score. Once you have been marked off for the milestone, continue the tasks below.

Task 2 - Reading from an XML File

In this task, you are going to create a PHP template which allows reading of XML data into our webpages.

1. Navigate to the virtual host you created in Lab 6, i.e., the `seng1050` directory.
Create a new directory called `data`.
In the `data` directory, create two new directories called `images` and `xml`.
2. Download the `airports.xml` file from Canvas (in the Week 8 module) and place it in the `xml` directory.
3. Download an any image you like that is related to an airport.
Rename it as `airport` while keeping the extension and place it in the `images` directory.
4. Within the `theme` directory, create a new file called `xml-template.php` and copy the following code into it:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title><?php echo $title; ?></title>
    <link rel="stylesheet" type="text/css" href="<?php echo $themefolder ?>/style.css" >
  </head>
  <body>
    <?php
      $xmlpath = $datafolder . "/xml/" . $ GET["name"] . ".xml";
      $root = simplexml_load_file($xmlpath);
      echo "<h1>" . $root->getName() . "</h1>";
      foreach ($root->children() as $children) {
        if ($children['code']) {
          echo "<p>Airport Code: " . $children['code'] . "</p>";
        }
        echo "<p>Airport Name: " . $children->xpath("name")[0] . "</p>";
      }
    ?>
  </body>
</html>
```

5. Return to the `seng1050` directory, create a new file called `xmltest.php` and copy the following code into it:

```
<?php
  $title = 'First PHP Webpage';
  $datafolder = 'data';
  $themefolder = 'theme';
  $content = '';
?>

<?php include($themefolder . '/xml-template.php'); ?>
```

6. Open your browser and type `http://seng1050.com/xmltest.php?name=airports` in the URL bar.
Now, you are able to view the contents of `airports.xml` according to what is defined by `xml-template.php`.

Before moving to the next step, try to think of the following:

- In the URL, why do we access `xmltest.php`? Why do we set the value of the attribute `name` to `airports`?
 - Why did we see the name of the root tag when the function `'getName()'` is called?
 - How `xml-template.php` defines the webpage output?
7. Edit your `xml-template.php` file again to retrieve the following information from `airports.xml` (*Hint: Refer to slides 29-32, 42-46, and 48 of Lecture 7*):

- The description of each airport should be displayed.
 - If a URL element exists and is not blank, this should be inserted into the `href` attribute of an `<a>` tag on the page.
 - If an image element exists and is not blank, this should be inserted into the `src` attribute of an `<img>` tag on the page.
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Task 3 - Writing to an XML file

If you have extra time at the end, try out the following tasks (useful for the bonus task of Assessment 3):

1. Modify `index.php` in the `seng1050` directory to create a new webpage containing a form with fields for 'name', 'description', and 'url'.

Note that these fields correspond to the XML elements in `airports.xml`.

Have this form submit as a POST request to a PHP script called `submit.php` that is in your `seng1050` directory.

2. Create the PHP script called `submit.php` in your `seng1050` directory.

Write the script so that it receives the values of the 'name', 'description', and 'url' fields from the form using POST.

After reading, it will check that the values for 'name', 'description' and 'url' are not empty, and add them to the file `airports.xml` as a new `airport` element (*Hint: Refer to slide 52 of Lecture 7*).
